

A Generalisation of Goursat’s Algorithm for Integration in Finite Terms

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Abstract

We give a self-contained, modern exposition of Édouard Goursat’s 1887 theorem on pseudo-elliptic integrals — those integrals of the form $\int F(t) dt/\sqrt{R(t)}$ with R a cubic or quartic polynomial that, despite living on a genus-1 algebraic curve, admit elementary antiderivatives. After reviewing integration in finite terms and Liouville’s theorem, we present Goursat’s two main theorems with proofs phrased in the language of Möbius automorphisms of the underlying hyperelliptic curve. We then develop a cube-root analog: for integrals of the form $\int F(t) dt/\sqrt[3]{R(t)}$ with R cubic, an order-3 Möbius substitution cyclically permuting the roots of R induces an eigendecomposition into three pieces. Two of the three eigenpieces (eigenvalues 1 and ω^2 , where $\omega = e^{2\pi i/3}$) descend through a chain of substitutions to genus-0 curves and yield elementary antiderivatives; the middle eigenpiece (eigenvalue ω) descends only to the genus-1 curve $y^3 = x(x - K)$ and is generically transcendental.

1 Introduction

Among the very first surprises of integral calculus is the asymmetry between

$$\int \frac{dt}{\sqrt{1-t^2}} = \arcsin(t) \quad \text{and} \quad \int \frac{dt}{\sqrt{(1-t^2)(1-k^2t^2)}} = ?$$

The first integral is elementary; the second is not. Both integrands look formally similar — a rational function divided by the square root of a polynomial — yet only the first admits a closed-form antiderivative in terms of the standard library of elementary functions. The deep reason is geometric: the curve $y^2 = 1 - t^2$ has *genus zero* (it is the unit circle), while the curve $y^2 = (1 - t^2)(1 - k^2t^2)$ has *genus one*, and on a curve of positive genus most 1-forms have no elementary antiderivative. The classical incomplete elliptic integral on the right is the prototype.

This raises a refined question: can a 1-form on a positive-genus curve *ever* integrate elementarily? The answer is yes, but only for very special integrands. The *pseudo-elliptic integrals* are precisely those of the form $\int F(t) dt/\sqrt{R(t)}$ which look elliptic but evaluate elementarily. Their existence is non-obvious, and producing systematic methods for detecting and reducing them was a recurrent theme in nineteenth-century analysis. Édouard Goursat’s note of 1887 [8] is the most elegant general account.

The purpose of this paper is twofold. First, in Sections 2–3, we give a self-contained modern exposition of Goursat’s two main theorems. The framework is standard differential algebra: an integral is elementary precisely when it satisfies the structural conclusion of Liouville’s theorem [15], and Goursat’s hypothesis turns out to be a clean Galois-theoretic condition imposed by the Möbius automorphisms of the curve $y^2 = R(t)$. Second, in Sections 4–5, we develop a cube-root analog of Goursat’s theory. The natural setting is the curve $y^3 = R(t)$, and we prove an analog of Goursat’s first theorem in which the binary anti-invariance condition

$F(t) + F(S(t)) = 0$ is replaced by an eigenvalue condition under an order-3 Möbius substitution.

The cube-root case exhibits a structural feature with no analog in the square-root world: of the three eigenspaces under a cyclic-Möbius action, two lead to elementary antiderivatives but the middle one stalls at a *cube-root elliptic curve* $y^3 = x(x - K)$. This curve also has genus one, and its standard holomorphic differential is generically transcendental. The result is a clean sufficient condition for a cube-root integral to be elementary, together with a precise description of the obstruction when the condition fails.

The possibility of a cube-root extension of Goursat’s procedure was hinted at over many years by Martin Welz in extensive postings on `sci.math.symbolic` [18], where particular cube-root integrands were observed to behave in a Goursat-like manner under cyclic Möbius substitutions; the present paper gives a complete formulation, proof, and algorithmic implementation of such an extension.

The structure of the paper is as follows. Section 2 reviews integration in finite terms and states Liouville’s principle in the form we will use. Section 3 is the modern presentation of Goursat: Theorem 3.3 (single-involution) and Theorem 3.5 (Klein four-group sum). Section 4 develops the cube-root analog: setup, statement of Theorem 4.3, proof, and a discussion of the middle-eigenvalue obstruction. Section 5 works through six representative examples. Section 6 closes with remarks on extensions and the relation to differential Galois theory.

2 Integration in finite terms and Liouville’s theorem

The framework for making “elementary integration” precise is the differential algebra of Joseph Liouville, refined by Ritt [14] and Rosenlicht [15], and given its modern algorithmic formulation through the work of Risch [12, 13], Davenport [7], Trager [16], and Bronstein [4].

Definition 2.1. A *differential field* is a field K of characteristic zero equipped with a derivation, i.e. a map $D: K \rightarrow K$ satisfying $D(a+b) = Da + Db$ and $D(ab) = aDb + bDa$. We write $a' = Da$ throughout.

The basic example is $K = \mathbb{C}(t)$ with $D = d/dt$. We will also consider algebraic extensions $K \subseteq L$, where the derivation extends uniquely.

Definition 2.2. Let $K \subseteq L$ be differential fields. We say L is an *elementary extension* of K if there is a tower

$$K = K_0 \subseteq K_1 \subseteq K_2 \subseteq \cdots \subseteq K_n = L$$

such that each $K_{i+1} = K_i(\theta_i)$ where one of the following holds:

- (i) θ_i is *algebraic* over K_i ;
- (ii) θ_i is an *exponential*: $\theta'_i/\theta_i \in K_i$;
- (iii) θ_i is a *logarithm*: $\theta'_i = \eta'/\eta$ for some $\eta \in K_i$.

An element $f \in K$ is said to have an *elementary integral* (or *integrate in finite terms*) if there exists an elementary extension $L \supseteq K$ and an element $g \in L$ with $g' = f$.

The class of elementary functions in the calculus textbook sense corresponds to elementary extensions of $\mathbb{C}(t)$.

Theorem 2.3 (Liouville’s Theorem; [15]). *Let K be a differential field with field of constants $\mathbb{C}$. If $f \in K$ has an elementary integral, then there exist $v \in K$, constants $c_1, \dots, c_n \in \mathbb{C}$, and elements $u_1, \dots, u_n \in K$ such that*

$$f = v' + \sum_{i=1}^n c_i \frac{u_i'}{u_i}. \quad (1)$$

The remarkable feature of (1) is that no new transcendentals appear: although the antiderivative may live in an extension $L \supsetneq K$, the structural form of f is dictated by elements of K itself. This is the precise statement that “the integral, if elementary, is the derivative of a rational function plus a sum of constant multiples of logarithms of rational functions.”

For our purposes we will need to apply Liouville’s theorem when K is an algebraic — specifically, a finite radical — extension of a rational function field. Throughout the rest of the paper we restrict attention to this purely algebraic setting, ignoring any further transcendental extensions: the integrands of interest are of the form $F(t)/\sqrt{R(t)}$ or $F(t)/\sqrt[3]{R(t)}$, and the relevant differential field is

$$K = \mathbb{C}(t, \theta), \quad \theta^n = R(t), \quad n \in \{2, 3\}.$$

Theorem 2.4 (Liouville’s Theorem in the algebraic case). *Let $K = \mathbb{C}(t)(\theta)$ where θ is algebraic over $\mathbb{C}(t)$, and let $f \in K$. If $\int f \, dt$ is elementary, then it can be written*

$$\int f \, dt = v + \sum_{i=1}^m c_i \log u_i$$

for some $v, u_1, \dots, u_m \in K$ and constants $c_i \in \mathbb{C}$.

This is the form of Liouville’s principle relevant to the integration of algebraic functions in the present paper. As an immediate consequence, if X is a smooth projective curve of genus $g \geq 1$ and ω is a holomorphic 1-form on X (a so-called *differential of the first kind*), then the function $P \mapsto \int_{P_0}^P \omega$ is *never* elementary as a function on X [4, 16]. More generally, the same conclusion holds for any non-zero *differential of the second kind* on X — a meromorphic 1-form with zero residue at every pole. The reason is structural: by Theorem 2.4, an elementary primitive $\int \omega$ on X would express ω as $dv + \sum c_i du_i/u_i$ for $v, u_i \in \mathbb{C}(X)$, and the logarithmic part contributes only third-kind differentials (poles with non-zero residues); a second-kind differential is therefore elementary iff exact, iff its periods on X vanish along all closed cycles. The space of second-kind differentials modulo exact ones on X has dimension g (it is one of the two Lagrangian halves of $H_{\text{dR}}^1(X)$, dual under the period pairing to the holomorphic differentials [4, Ch. 5]), so a non-zero second-kind differential on a positive-genus curve is never exact, hence never has an elementary primitive. This refines the classical fact that $\int dt/\sqrt{(1-t^2)(1-k^2t^2)}$ is not elementary, and we will rely on it in Section 4 to identify the obstruction in the cube-root case.

Remark 2.5. The constructive content of Theorem 2.4 is the algebraic case of the Risch–Trager–Bronstein algorithm [12, 13, 7, 16, 4]: given $f \in K$ as input, the algorithm decides in finite time whether $\int f \, dt$ is elementary, and if so produces it. Goursat’s theorems below predate these developments by nearly a century and provide closed-form reductions in cases where the algebraic structure is particularly transparent.

3 Goursat’s pseudo-elliptic theorem

Throughout this section, $R(t) \in \mathbb{C}[t]$ denotes a polynomial of degree 3 or 4 with *simple roots* — i.e., all roots distinct, or equivalently $\gcd(R, R') = 1$. We write $C := \{(t, y) \in \overline{\mathbb{C}}^2 : y^2 = R(t)\}$

for the smooth projective completion of the affine curve $y^2 = R(t)$; this is a curve of genus 1. We are interested in the integrals

$$I = \int F(t) \frac{dt}{\sqrt{R(t)}} \quad (2)$$

with $F \in \mathbb{C}(t)$.

Definition 3.1. The integral (2) is called *pseudo-elliptic* if it is elementary in the sense of Definition 2.2, i.e. if it admits a closed-form antiderivative built from rational functions, $\sqrt{R}$, exponentials, and logarithms.

The reason for the term *pseudo-elliptic* is that the integrand lives most naturally as a 1-form on C , and a generic such integral is genuinely transcendental (an elliptic integral). The pseudo-elliptic ones are the rare exceptions where the integral happens to be elementary.

3.1 The role of Möbius involutions

The key idea is to exploit symmetries of the curve C . The Möbius group $\mathrm{PGL}_2(\mathbb{C})$ acts on $\mathbb{C} \cup \{\infty\} = \mathbb{P}^1(\mathbb{C})$ by fractional linear transformations $S(t) = (At + B)/(Ct + D)$. An element $S \in \mathrm{PGL}_2(\mathbb{C})$ is an *involution* if $S^2 = \mathrm{id}$ and $S \neq \mathrm{id}$. A non-trivial Möbius involution has exactly two fixed points on $\mathbb{P}^1(\mathbb{C})$.

Lemma 3.2. *A non-trivial Möbius involution S on $\mathbb{P}^1(\mathbb{C})$ permuting the roots of a quartic R with simple roots must permute them in two pairs (no S -fixed root).*

If the four roots are $\{a, b, c, d\}$ and S swaps $\{a, b\}$ with $\{c, d\}$, then explicitly

$$S(t) = \frac{(ab - cd)t + (a + b)cd - (c + d)ab}{[(a + b) - (c + d)]t - (ab - cd)}. \quad (3)$$

There are exactly three such involutions, corresponding to the three ways of pairing four points; together with the identity they form a Klein four-group $V_4 \subseteq \mathrm{PGL}_2(\mathbb{C})$.

Proof. A Möbius transformation is determined by its action on three points; once the values $S(a) = b$, $S(b) = a$, $S(c) = d$ are fixed (and consistency requires $S(d) = c$), the transformation is uniquely determined and one verifies (3) by direct computation. The product of any two of these three involutions is the third (as one checks on the pairings), so they generate a V_4 subgroup. $\square$

3.2 Goursat's first theorem

Theorem 3.3 (Goursat [8], Theorem 1). *Let $R \in \mathbb{C}[t]$ be a polynomial of degree 3 or 4 with simple roots, and let $S \in \mathrm{PGL}_2(\mathbb{C})$ be a non-trivial Möbius involution permuting the roots of R in pairs. If $F \in \mathbb{C}(t)$ satisfies the anti-invariance condition*

$$F(t) + F(S(t)) = 0 \quad (\text{identically}), \quad (4)$$

then the integral $\int F(t) dt / \sqrt{R(t)}$ is pseudo-elliptic. Explicitly, let $\alpha, \beta \in \mathbb{P}^1(\mathbb{C})$ be the two fixed points of S , and set

$$u = \frac{t - \alpha}{t - \beta}, \quad x = u^2.$$

Then

$$\int F(t) \frac{dt}{\sqrt{R(t)}} = \int \frac{G(x) dx}{2\sqrt{Q(x)}}$$

for some $G \in \mathbb{C}(x)$ and quadratic $Q \in \mathbb{C}[x]$, the latter integral being elementary.

Proof. We argue in five steps.

Step 1 (canonical form of S). Since S has order 2 with two distinct fixed points α, β , the Möbius transformation $u = (t - \alpha)/(t - \beta)$ sends $\alpha \mapsto 0$, $\beta \mapsto \infty$, and conjugates S into a Möbius transformation of u fixing $\{0, \infty\}$. Such a transformation is multiplication by a scalar λ , and the involution condition $\lambda^2 = 1$ together with $\lambda \neq 1$ forces $\lambda = -1$. Thus in u -coordinates, S acts as $u \mapsto -u$. We write $t(u) = (\alpha - \beta u)/(1 - u)$ for the inverse change of variable.

Step 2 (transformation of R). Each linear factor $t - r_i$ becomes

$$t(u) - r_i = \frac{(\alpha - r_i) - u(\beta - r_i)}{1 - u}.$$

Hence (assuming R is quartic; the cubic case is identical with $r_4 = \infty$ and one factor reading 1)

$$R(t(u))(1 - u)^4 = \prod_{i=1}^4 [(\alpha - r_i) - u(\beta - r_i)].$$

The roots u_i of this polynomial in u are $u_i = (\alpha - r_i)/(\beta - r_i) = -(r_i - \alpha)/(\beta - r_i) \dots$, more usefully, the image of r_i under $r \mapsto (r - \alpha)/(r - \beta)$. Since S swaps the roots r_i in pairs and acts as $u \mapsto -u$, the u_i come in pairs $\{u_1, -u_1\}$ and $\{u_3, -u_3\}$. Therefore

$$R(t(u))(1 - u)^4 = C(u^2 - u_1^2)(u^2 - u_3^2) \quad (5)$$

for some non-zero constant C . The right-hand side is a polynomial in u^2 .

Step 3 (transformation of the differential). A direct computation gives

$$dt = \frac{\alpha - \beta}{(1 - u)^2} du.$$

Combined with (5) this yields

$$\frac{dt}{\sqrt{R(t)}} = \frac{(\alpha - \beta) du / (1 - u)^2}{\sqrt{C(u^2 - u_1^2)(u^2 - u_3^2) / (1 - u)^2}} = \frac{(\alpha - \beta) du}{\sqrt{C(u^2 - u_1^2)(u^2 - u_3^2)}}. \quad (6)$$

The factors $(1 - u)^2$ from the Jacobian and from the radical exactly cancel.

Step 4 (oddness of F in u). Anti-invariance (4) reads, in u -coordinates, $F(t(-u)) = F(S(t(u))) = -F(t(u))$. Hence $u \mapsto F(t(u))$ is an odd rational function of u , and admits a representation

$$F(t(u)) = u G(u^2)$$

for some $G \in \mathbb{C}(x)$.

Step 5 (descent via $x = u^2$). Combining the previous steps:

$$\int F(t) \frac{dt}{\sqrt{R(t)}} = (\alpha - \beta) \int \frac{u G(u^2) du}{\sqrt{C(u^2 - u_1^2)(u^2 - u_3^2)}}.$$

Substituting $x = u^2$, $dx = 2u du$:

$$= \frac{\alpha - \beta}{2\sqrt{C}} \int \frac{G(x) dx}{\sqrt{(x - u_1^2)(x - u_3^2)}}.$$

The radical is now $\sqrt{Q(x)}$ for the quadratic $Q(x) = (x - u_1^2)(x - u_3^2)$. The integral $\int G(x) dx / \sqrt{Q(x)}$ with G rational and Q quadratic is elementary by classical methods (partial fractions of G together with Euler substitution). This completes the proof. $\square$

Remark 3.4. The case where one of the fixed points is at infinity (so that $S(t) = 2\alpha - t$) is covered by the same argument, the substitution simplifying to $u = t - \alpha$.

3.3 Goursat's second theorem

The three involutions of Lemma 3.2 together with the identity form V_4 , and one can ask for a sum-of-orbits criterion that detects pseudo-ellipticity.

Theorem 3.5 (Goursat [8], Theorem 2). *Let $R \in \mathbb{C}[t]$ be a polynomial of degree 3 or 4 with simple roots, with Möbius involutions S_1, S_2, S_3 permuting the roots in pairs, and let $F \in \mathbb{C}(t)$. If*

$$F + F \circ S_1 + F \circ S_2 + F \circ S_3 = 0 \quad (7)$$

identically, then $\int F dt/\sqrt{R}$ is pseudo-elliptic.

Proof. The group $V_4 = \{I, S_1, S_2, S_3\}$ has four irreducible characters $\chi_0, \chi_1, \chi_2, \chi_3: V_4 \rightarrow \{\pm 1\}$, with χ_0 the trivial character. We label them so that $\chi_j(S_j) = +1$ for $j = 1, 2, 3$, and $\chi_j(S_k) = -1$ for $j \neq k$ in $\{1, 2, 3\}$.

The associated character projections decompose $\mathbb{C}(t)$ as a direct sum of χ_j -eigenspaces under the natural V_4 -action. Setting

$$F^{(j)} := \frac{1}{4} \sum_{g \in V_4} \chi_j(g) F \circ g,$$

hypothesis (7) says exactly $F^{(0)} = 0$, hence $F = F^{(1)} + F^{(2)} + F^{(3)}$.

We claim $F^{(j)}$ is anti-invariant under S_k for every $k \neq j$ (and invariant under S_j , as is automatic). Indeed, applying S_k to $F^{(j)}$ permutes the summands by the action $g \mapsto gS_k$, which on characters acts by $\chi_j(g) \mapsto \chi_j(gS_k) = \chi_j(g)\chi_j(S_k)$. For $k \neq j$ this multiplies the projection by -1 , giving anti-invariance.

By Theorem 3.3, applied to $F^{(j)}$ with any involution S_k , $k \neq j$, each integral $\int F^{(j)} dt/\sqrt{R}$ is pseudo-elliptic. Summing the three antiderivatives yields an elementary expression for $\int F dt/\sqrt{R}$. $\square$

Remark 3.6. The character labelling above corrects a common indexing issue. Goursat's original argument writes the projections as

$$\begin{aligned} F^{(1)} &= (F - F \circ S_1 - F \circ S_2 + F \circ S_3)/4, \\ F^{(2)} &= (F - F \circ S_1 + F \circ S_2 - F \circ S_3)/4, \\ F^{(3)} &= (F + F \circ S_1 - F \circ S_2 - F \circ S_3)/4, \end{aligned}$$

and one must check carefully which involution S_k to pair with each $F^{(j)}$ when invoking Theorem 3.3: $F^{(j)}$ is anti-invariant under exactly the two involutions S_k with $k \neq j$, hence either of those two works.

3.4 Geometric interpretation

Although Goursat does not phrase his theorems in this way, the modern reading is illuminating. Each Möbius involution S of $\mathbb{P}^1$ permuting the roots of R extends to an involution $\tilde{S}$ of the curve $C: y^2 = R(t)$ that sends y to $\pm y \cdot J_S(t)$ for an explicit Jacobian factor J_S . The pseudo-elliptic condition is then the statement that the differential $F dt/y$ is a (-1) -eigenvector of $\tilde{S}^*$ acting on Ω_C^1 . Theorem 3.5's sum criterion is the orthogonality of $F dt/y$ to the trivial-character isotypic component of Ω_C^1 under the V_4 -action — so all of $F dt/y$ lives in non-trivial isotypics, each of which is an eigenspace of *some* involution.

We will see in the next section that the analogous picture for cube radicals is more subtle: there are three eigenspaces under a $\mathbb{Z}/3$ -action, but only two of them yield elementary integrals.

4 The cube-root analog

We now consider integrands of the form

$$J = \int F(t) \frac{dt}{\sqrt[3]{R(t)}}, \quad F \in \mathbb{C}(t), \quad R \in \mathbb{C}[t]. \quad (8)$$

The natural geometric setting is the cyclic cubic cover $X_R: y^3 = R(t)$.

4.1 Genus arithmetic

A direct application of Riemann–Hurwitz to the projection $X_R \rightarrow \mathbb{P}^1$, $(t, y) \mapsto t$, of degree 3 yields the genus of X_R for R of degree n with simple roots:

$$g(X_R) = \begin{cases} n-1 & \text{if } n \not\equiv 0 \pmod{3}, \\ n-2 & \text{if } n \equiv 0 \pmod{3}. \end{cases}$$

The case relevant to us is genus 1:

$\deg R$	1	2	3	4	5
$g(X_R)$	0	1	1	3	4

We will focus on the genus-1 cases $\deg R \in \{2, 3\}$. The case $\deg R = 3$ is the cleanest analog of Goursat’s quartic-square-root setting (also genus 1), and we treat it first; the case $\deg R = 2$ may be reduced to $\deg R = 3$ by viewing one root as lying at infinity.

4.2 Order-3 Möbius substitutions

Suppose R is cubic with simple roots $\{a, b, c\} \subset \mathbb{C}$.

Lemma 4.1. *There exists a unique Möbius transformation $S \in \text{PGL}_2(\mathbb{C})$ of order exactly 3 sending $a \mapsto b \mapsto c \mapsto a$. Its inverse $S^{-1} = S^2$ sends $a \mapsto c \mapsto b \mapsto a$. The two fixed points $\alpha, \beta \in \mathbb{P}^1(\mathbb{C})$ of S are not among $\{a, b, c\}$.*

Proof. A Möbius transformation is determined by its action on three points, so S exists and is unique. To see S has order exactly 3: it is non-trivial since $S(a) = b \neq a$, and $S^3(a) = S^2(b) = S(c) = a$ together with the analogous identities on b, c show that S^3 fixes three distinct points $\{a, b, c\}$, hence $S^3 = \text{id}$. So $\text{ord}(S)$ divides 3, and is therefore exactly 3.

For the fixed points: if S fixed some $r \in \{a, b, c\}$, then $S(r) = r$ would contradict the cyclic permutation $a \mapsto b \mapsto c \mapsto a$ (which has no fixed point in $\{a, b, c\}$). $\square$

In coordinates adapted to S , the cyclic action takes a particularly simple form.

Lemma 4.2. *Let α, β be the fixed points of S in Lemma 4.1, and set*

$$z := \frac{t - \alpha}{t - \beta}.$$

Then S acts in z -coordinates as $z \mapsto \omega z$, where $\omega = e^{2\pi i/3}$, and there exist non-zero constants $c, K \in \mathbb{C}$ with

$$R(t(z)) (1 - z)^3 = c(z^3 - K). \quad (9)$$

Proof. The substitution $z = (t - \alpha)/(t - \beta)$ sends $\alpha \mapsto 0$ and $\beta \mapsto \infty$, conjugating S to a Möbius transformation of z fixing $\{0, \infty\}$, hence multiplication by a scalar λ . The order condition $\lambda^3 = 1$ together with $\lambda \neq 1$ forces $\lambda \in \{\omega, \omega^2\}$; relabelling $S \leftrightarrow S^2$ if necessary, we take $\lambda = \omega$.

The inverse change of variable is $t(z) = (\alpha - \beta z)/(1 - z)$. As in Step 2 of the proof of Theorem 3.3, multiplying through by $(1 - z)^3$ converts $R(t(z))$ into a polynomial of degree 3 in z . Its roots in z -coordinates are the images of a, b, c under $r \mapsto (r - \alpha)/(r - \beta)$. Because S acts as multiplication by ω and permutes $\{a, b, c\}$ cyclically, those images form a single ω -orbit $\{\zeta, \omega\zeta, \omega^2\zeta\}$. Hence

$$R(t(z))(1 - z)^3 = c(z - \zeta)(z - \omega\zeta)(z - \omega^2\zeta) = c(z^3 - \zeta^3),$$

and $K = \zeta^3$. □

4.3 The eigendecomposition

In z -coordinates the cube-radical differential takes a particularly simple form. Writing $t = t(z) = (\alpha - \beta z)/(1 - z)$, $dt = (\alpha - \beta) dz/(1 - z)^2$, and using (9):

$$\frac{F(t) dt}{R(t)^{1/3}} = \frac{(\alpha - \beta) F(t(z)) dz / (1 - z)^2}{c^{1/3}(z^3 - K)^{1/3}/(1 - z)} = H(z) \frac{dz}{(z^3 - K)^{1/3}}, \quad (10)$$

where

$$H(z) := \frac{(\alpha - \beta) F(t(z))}{c^{1/3} (1 - z)}. \quad (11)$$

Here we have chosen a specific cube root $c^{1/3}$ once and for all; different choices alter H and the radical $(z^3 - K)^{1/3}$ by mutually cancelling third roots of unity.

The substitution $\sigma: z \mapsto \omega z$ generates a cyclic group of order 3 acting on $\mathbb{C}(z)$. Under this action, $\mathbb{C}(z)$ decomposes as a direct sum of three eigenspaces:

$$\mathbb{C}(z) = V_0 \oplus V_1 \oplus V_2, \quad V_k = \{f \in \mathbb{C}(z) : f(\omega z) = \omega^k f(z)\}.$$

The eigenspaces are characterized by

$$V_0 = \mathbb{C}(z^3), \quad V_1 = z \cdot \mathbb{C}(z^3), \quad V_2 = z^2 \cdot \mathbb{C}(z^3),$$

and the projection onto V_k is

$$P_k f = \frac{1}{3} (f(z) + \omega^{-k} f(\omega z) + \omega^{-2k} f(\omega^2 z)). \quad (12)$$

We can therefore write $H = H_0 + H_1 + H_2$ uniquely with $H_k = P_k H \in V_k$, and each piece contributes a separate integral

$$J_k := \int H_k(z) \frac{dz}{(z^3 - K)^{1/3}}.$$

The integrals J_0, J_1, J_2 behave very differently from one another, and this is the main phenomenon.

4.4 The main theorem

Theorem 4.3 (Cube-root pseudo-elementarity). *Let $R \in \mathbb{C}[t]$ be a cubic polynomial with simple roots, let $S \in \text{PGL}_2(\mathbb{C})$ be the order-3 Möbius transformation cyclically permuting the roots of R , and let $H \in \mathbb{C}(z)$ be the function defined by (11) for $F \in \mathbb{C}(t)$. Decompose $H = H_0 + H_1 + H_2$ as above, and write $H_k(z) = z^k \varphi_k(z^3)$ for unique rational functions $\varphi_k \in \mathbb{C}(x)$.*

- (i) **(Sufficient criterion for elementarity.)** *If $H_1 \equiv 0$, then $\int F \, dt / \sqrt[3]{R}$ is elementary, with explicit antiderivative obtained from the rational reductions of J_0 and J_2 on the genus-zero curves $Y_0: y^3 = x^2(x - K)$ and $Y_2: y^3 = x - K$ respectively.*
- (ii) **(Sufficient criterion for non-elementarity.)** *If $H_1 \not\equiv 0$ and the descended differential $\omega_1 = \varphi_1(x) \, dx / \sqrt[3]{x(x - K)}$ on the genus-one curve $Y_1: y^3 = x(x - K)$ is of the second kind (has zero residue at every pole on Y_1), then $\int F \, dt / \sqrt[3]{R}$ is not elementary.*
- (iii) **(A common sufficient condition for (ii).)** *If $\varphi_1 \in \mathbb{C}[x]$ is a polynomial (in particular, this holds whenever $F \in \mathbb{C}[t]$ is a polynomial), then ω_1 is automatically of the second kind on Y_1 , so the criterion in (ii) reduces to the test $H_1 \not\equiv 0$.*

Proof. Substituting $x = z^3$ into each $J_k = \int H_k(z) \, dz / (z^3 - K)^{1/3}$ and using $dx = 3z^2 \, dz$:

$$J_0 = \frac{1}{3} \int \frac{\varphi_0(x) \, dx}{x^{2/3}(x - K)^{1/3}}, \quad J_1 = \frac{1}{3} \int \frac{\varphi_1(x) \, dx}{x^{1/3}(x - K)^{1/3}}, \quad J_2 = \frac{1}{3} \int \frac{\varphi_2(x) \, dx}{(x - K)^{1/3}}.$$

The radicands $x^2(x - K)$, $x(x - K)$, $x - K$ define curves Y_0, Y_1, Y_2 respectively, and we treat each.

J_2 is elementary. The radical $(x - K)^{1/3}$ defines a curve isomorphic to the affine line via $x = K + w^3$, after which $dx = 3w^2 \, dw$ and

$$J_2 = \int w \varphi_2(K + w^3) \, dw,$$

which is the integral of a rational function in w and hence elementary.

J_0 is elementary. The radicand $x^2(x - K)$ has a *double* root at $x = 0$ and a *simple* root at $x = K$. The cube-radical curve $Y_0: y^3 = x^2(x - K)$ has genus zero, parametrized rationally by

$$x = \frac{K}{1 - w^3}, \quad y = \frac{Kw}{1 - w^3}, \quad (13)$$

giving $dx/y = 3w \, dw / (1 - w^3)$. Therefore

$$J_0 = \int \frac{\varphi_0(K/(1 - w^3)) w}{1 - w^3} \, dw,$$

the integral of a rational function in w , hence elementary.

This proves (i): if $H_1 \equiv 0$ then $J_0 + J_2$ furnishes an elementary antiderivative.

Proof of (ii). Assume $H_1 \not\equiv 0$ and that ω_1 is of the second kind on Y_1 . The curve $Y_1: y^3 = x(x - K)$ is the smooth projective cube-radical curve of $x(x - K)$, which has genus 1 (Riemann–Hurwitz: degree-3 cover of $\mathbb{P}^1$ ramified at $0, K, \infty$, each with $e = 3$, gives $2g - 2 = -6 + 3 \cdot 2 = 0$).

A non-zero meromorphic differential ω on a smooth projective curve C has an elementary primitive in the function field $\mathbb{C}(C)$ if and only if (Risch, Trager, Bronstein [16, 4])

$$\omega = df + \sum_i c_i \frac{dg_i}{g_i}$$

for some $f, g_i \in \mathbb{C}(C)$ and constants $c_i \in \mathbb{C}$. Each summand $c_i dg_i/g_i$ is a third-kind differential whose poles are simple with integer-multiple-of- c_i residues. Subtracting the logarithmic part, the remainder $\omega - \sum c_i dg_i/g_i$ is of the second kind (zero residues at all poles). If the original ω was already of the second kind, then *either* $\omega = df$ (exact) or no decomposition exists.

For a curve of positive genus, an exact second-kind differential vanishes identically as a class in $H_{\text{dR}}^1(C)/d\mathbb{C}(C)$. Concretely, $\omega = df$ for $f \in \mathbb{C}(C)$ implies that the periods $\oint_\gamma \omega$ vanish along every closed cycle γ on C , which in turn forces ω to vanish identically when no non-zero exact second-kind differential exists on C . By a classical theorem (see, e.g., [4, Ch. 5]), the space of second-kind differentials modulo exact differentials on a curve of genus g has dimension g , hence is non-zero whenever $g \geq 1$.

Therefore: if ω_1 is a non-zero second-kind differential on Y_1 , then $\int \omega_1$ is not elementary as a function on Y_1 , and consequently J_1 (which equals $\frac{1}{3} \int \omega_1$) is not elementary either. Together with (i), this gives (ii).

Proof of (iii). We verify that $\omega_1 = \varphi_1(x) dx/y$ on Y_1 (where $y = (x(x - K))^{1/3}$) is of the second kind whenever φ_1 is a polynomial. The branch points are at $x = 0, K, \infty$, all with full ramification ($e = 3$).

At $x = 0$: a local uniformizer τ on Y_1 satisfies $x = \tau^3$ (with $y = (x(x - K))^{1/3} = \tau \cdot (-K)^{1/3}(1 + O(\tau^3))$). Thus $dx/y = 3\tau^2 d\tau/[\tau(-K)^{1/3}(1 + \dots)] = (3/(-K)^{1/3})(1 + \dots) d\tau$, so dx/y has a simple zero at P_0 . Multiplying by $\varphi_1(\tau^3) = \varphi_1(0) + O(\tau^3)$ gives a regular differential at P_0 for any polynomial φ_1 .

At $x = K$: symmetric analysis gives a simple zero of $\varphi_1(x) dx/y$ at P_K .

At $x = \infty$: a local uniformizer τ satisfies $x = 1/\tau^3$, $y = 1/\tau^2 \cdot (1 - K\tau^3)^{1/3} = 1/\tau^2 \cdot (1 + O(\tau^3))$. So $dx/y = (-3/\tau^4) d\tau/(1/\tau^2)(1 + \dots) = -3\tau^{-2}(1 + O(\tau^3)) d\tau$, a pole of order 2 at P_∞ . For polynomial $\varphi_1(x) = c_d x^d + \dots$ ($d \geq 0$), $\varphi_1(x) = c_d/\tau^{3d} + \dots$, so $\varphi_1 dx/y$ has a pole of order $3d + 2$ at P_∞ . The coefficient of τ^{-1} in the Laurent expansion (the residue) requires $3d + 2 \equiv 1 \pmod{?}$, but since $3d + 2 \geq 2$ for all $d \geq 0$, the only way to get a τ^{-1} term is $3d + 2 = 1$, which has no non-negative integer solution. Hence the residue at P_∞ is zero.

Thus ω_1 has zero residue at every point of Y_1 , i.e., is of the second kind, completing (iii). $\square$

Remark 4.4. The polynomial- φ_1 hypothesis in (iii) is satisfied whenever F is a polynomial in t . Indeed, if $F \in \mathbb{C}[t]$, then in the canonical coordinates $z = (t - \alpha)/(t - \beta)$ (or $z = t - \alpha$ when $\beta = \infty$), the function $F(t(z))$ is rational in z with poles only at $z = 1$ (from the Möbius substitution; absent when $\beta = \infty$). The factor $(\alpha - \beta)/(c^{1/3}(1 - z))$ in the definition of H contributes one more pole at $z = 1$. After eigenspace projection, $H_1(z)$ is a rational function whose denominator divides a power of $(1 - z^3) = (1 - z)(1 + z + z^2)$; equivalently, $\varphi_1(x)$ has poles only at $x = 1$ (which is mapped to $z = 1$ via $z^3 = x$). This pole, however, sits at the Möbius image of the second fixed point β , which is generically *not* a root of R and so corresponds to a non-branch point on Y_1 only if $\beta = \infty$ (in which case $z = 1$ has no preimage in finite z and φ_1 is polynomial). For finite β , φ_1 may have a pole at $x = 1$, but the corresponding differential typically remains of the second kind because $z = 1$ projects to a branch point of Y_1 in many cases of interest. We illustrate both polynomial and rational- φ_1 scenarios in Section 5.

Remark 4.5. The dichotomy in Theorem 4.3 has *no* analog in the square-root case. Under a $\mathbb{Z}/2$ -action there are only two eigenvalues ± 1 ; the trivial-character component is killed by the orbit-sum (giving Theorem 3.5), and the unique non-trivial component descends elementarily (Theorem 3.3). For the $\mathbb{Z}/3$ -action there are three eigenvalues $1, \omega, \omega^2$, and the middle one is genuinely obstructive: it descends only to the cube-root analog of a Weierstrass elliptic curve, $y^3 = x(x - K)$.

Remark 4.6 (Degree-2 radicand). For R of degree 2, the cube-radical curve $y^3 = R(t)$ still has genus 1, and Theorem 4.3 applies without modification. The proof of Lemma 4.2 extends

uniformly: when $\deg R = 2$, only two factors $t(z) - r_i$ contribute and produce a $1/(1-z)^2$ rather than $1/(1-z)^3$, but multiplying through by $(1-z)^3$ still gives a polynomial of degree 3 in z . The "extra" factor $(1-z)$ encodes the implicit ramification point at ∞ of the projection $X_R \rightarrow \mathbb{P}^1$.

Two structural features specific to $\deg R = 2$ deserve emphasis.

(a) $K = 1$ is forced. Because S cyclically permutes the two finite roots and the point at ∞ , the three roots in z -coordinates form a $\langle \omega \rangle$ -orbit on $\mathbb{P}^1(\mathbb{C})$ containing $z(\infty) = 1$, hence are exactly $\{1, \omega, \omega^2\}$. Therefore

$$R(t(z))(1-z)^3 = c(z^3 - 1).$$

For cubic R , by contrast, K depends on the configuration of the roots and may be any non-zero complex number.

(b) The fixed points of S are non-real complex conjugates (when $R \in \mathbb{R}[t]$). A direct computation shows that the unique cyclic Möbius transformation permuting $\{r_1, r_2, \infty\}$ as $r_1 \rightarrow r_2 \rightarrow \infty \rightarrow r_1$ is

$$S(t) = \frac{r_1 t - (r_1^2 - r_1 r_2 + r_2^2)}{t - r_2},$$

with fixed-point equation $t^2 - (r_1 + r_2)t + (r_1^2 - r_1 r_2 + r_2^2) = 0$ and discriminant $-3(r_1 - r_2)^2$. Since $r_1 \neq r_2$, the discriminant is strictly negative, hence

$$\alpha, \beta = \frac{r_1 + r_2}{2} \pm \frac{i\sqrt{3}}{2}(r_1 - r_2),$$

a pair of non-real complex conjugates. In particular, the change of variable $z = (t - \alpha)/(t - \beta)$ is unavoidably non-trivial — there is no degree-2 radicand for which it degenerates to $z = t - \alpha$, in contrast to the cubic case (e.g. $R = t^3 - 1$).

The two features above combine to force a uniform conclusion when $F = 1$:

Corollary 4.7 (Non-elementarity for quadratic radicand). *Let $R(t) \in \mathbb{C}[t]$ be a quadratic polynomial with two distinct roots. Then*

$$\int \frac{dt}{\sqrt[3]{R(t)}}$$

is not elementary.

Proof. Specialize Theorem 4.3 to $F = 1$. By (11),

$$H(z) = \frac{(\alpha - \beta) F(t(z))}{c^{1/3}(1-z)} = \frac{c_0}{1-z}, \quad c_0 := \frac{\alpha - \beta}{c^{1/3}}.$$

By Remark 4.6(b), both fixed points α, β are finite, so $\alpha - \beta \neq 0$ and hence $c_0 \neq 0$. Apply the geometric-series identity

$$\frac{1}{1-z} = \frac{1+z+z^2}{1-z^3}$$

to read off the eigendecomposition:

$$H_0 = \frac{c_0}{1-z^3}, \quad H_1 = \frac{c_0 z}{1-z^3}, \quad H_2 = \frac{c_0 z^2}{1-z^3}.$$

Since $c_0 \neq 0$, we have $H_1 \neq 0$. The associated rational function $\varphi_1(x) = c_0/(1-x)$ has a single pole at $x = 1$, but $K = 1$ in the canonical form for R quadratic (Remark 4.6(b)), so this pole sits at the branch point $x = K$ of Y_1 . We verify that the resulting differential

$$\omega_1 = \frac{\varphi_1(x) dx}{\sqrt[3]{x(x-K)}} = \frac{c_0 dx}{(1-x)\sqrt[3]{x(x-1)}}$$

is of the second kind on Y_1 . With local uniformizer τ at P_K given by $x - K = \tau^3$ (so $y = \tau \cdot K^{1/3}(1 + O(\tau^3))$), we have $\varphi_1(x) = -c_0/\tau^3$ and $dx/y = (3/K^{1/3})(1 + O(\tau^3)) d\tau$, giving $\omega_1 = (-3c_0/K^{1/3})\tau^{-3}(1 + O(\tau^3)) d\tau$. The Laurent expansion has no τ^{-1} term, so the residue at P_K is zero. At the other branch points P_0 and P_∞ , φ_1 is regular and the analysis of Theorem 4.3(iii) applies: the residues there are also zero. Therefore ω_1 is non-zero and of the second kind, and Theorem 4.3(ii) concludes that the integral is not elementary. $\square$

Remark 4.8. The contrast with the square-root case is striking: every integral $\int dt/\sqrt{R(t)}$ with $\deg R = 2$ is elementary (yielding arcsin, arccosh, or a logarithm depending on the sign pattern), because the curve $y^2 = R(t)$ has genus 0. The cube-radical curve $y^3 = R(t)$ has genus 1, and Corollary 4.7 shows that for $F = 1$ the obstruction in Theorem 4.3(ii) is generic — indeed, unavoidable. The point is that under the (unavoidable) substitution $z = (t - \alpha)/(t - \beta)$, the constant function $F(t) = 1$ does not remain constant: it acquires a pole at $z = 1$ (corresponding to $t = \infty$), and that single pole is enough to populate all three eigencomponents of H .

The same eigenspace machinery decides elementarity for the dual exponent $2/3$, and combining the two criteria handles the full algebraic field $\mathbb{C}(t, y)$ with $y^3 = R(t)$.

Corollary 4.9 (Cube-root analog at exponent $2/3$ and full algebraic field). *Let $R \in \mathbb{C}[t]$ be a cubic polynomial with simple roots, S the order-3 Möbius transformation of Lemma 4.1, α, β its fixed points, $z = (t - \alpha)/(t - \beta)$, and c, K the constants of Lemma 4.2. For $F \in \mathbb{C}(t)$, define*

$$\tilde{H}(z) := \frac{(\alpha - \beta) F(t(z))}{c^{2/3}} \quad (14)$$

(with the convention that the prefactor $\alpha - \beta$ is replaced by 1 when $\beta = \infty$), and decompose $\tilde{H} = \tilde{H}_0 + \tilde{H}_1 + \tilde{H}_2$ into ω -eigencomponents under $z \mapsto \omega z$, with $\tilde{H}_k(z) = z^k \tilde{\varphi}_k(z^3)$.

- (i) (Sufficient criterion for elementarity at $2/3$.) *If $\tilde{H}_0 \equiv 0$, then the integral $\int F(t) dt/R(t)^{2/3}$ is elementary, with the explicit reduction*

$$\int \frac{F(t) dt}{R(t)^{2/3}} = - \int \frac{\tilde{\varphi}_1\left(\frac{Ks^3}{s^3-1}\right) s ds}{s^3-1} + \int \tilde{\varphi}_2(u^3 + K) du, \quad (15)$$

both pieces being rational integrals in the parameter variables s and u .

- (ii) (Sufficient criterion for non-elementarity at $2/3$.) *If $\tilde{H}_0 \not\equiv 0$ and the descended differential $\tilde{\omega}_0 = \tilde{\varphi}_0(x) dx/[x^{2/3}(x - K)^{2/3}]$ on the curve $\tilde{Y}_0: y^3 = x^2(x - K)^2$ (birationally identified with $Y_1: v^3 = x(x - K)$ via $v^3 = x(x - K)$) is of the second kind, then the integral is not elementary. This holds in particular when $\tilde{\varphi}_0$ is a polynomial.*

- (iii) (Full algebraic field.) *Every $G \in \mathbb{C}(t, y)$ with $y^3 = R(t)$ admits a unique representation $G = G_0(t) + G_1(t)y + G_2(t)y^2$ with $G_i \in \mathbb{C}(t)$, and*

$$\int G dt = \int G_0 dt + \int \frac{G_2 R dt}{R^{1/3}} + \int \frac{G_1 R dt}{R^{2/3}}. \quad (16)$$

The first piece is rational; elementarity of the second is decided by Theorem 4.3 applied to $G_2 R$, and elementarity of the third by parts (i)–(ii) applied to $G_1 R$.

Proof. (i) Since $R(t)(1 - z)^3 = c(z^3 - K)$ by Lemma 4.2 and $dt = (\alpha - \beta) dz/(1 - z)^2$, one computes

$$R(t)^{2/3} = \frac{c^{2/3}(z^3 - K)^{2/3}}{(1 - z)^2} \implies \frac{dt}{R(t)^{2/3}} = \frac{(\alpha - \beta) dz}{c^{2/3}(z^3 - K)^{2/3}},$$

so the $(1 - z)^2$ factors cancel exactly — in contrast to (11) for the $1/3$ -exponent, no $1/(1 - z)$ factor remains. With $\tilde{H}$ as in (14),

$$\frac{F(t) dt}{R(t)^{2/3}} = \frac{\tilde{H}(z) dz}{(z^3 - K)^{2/3}}.$$

Substituting $x = z^3$ in each eigenvector gives

$$\int \frac{\tilde{H}_k(z) dz}{(z^3 - K)^{2/3}} = \tilde{J}_k := \int \frac{\tilde{\varphi}_k(x) dx}{3 x^{(2-k)/3} (x - K)^{2/3}}, \quad k \in \{0, 1, 2\},$$

so the integral lives on the curve $\tilde{Y}_k: y^3 = x^{2-k}(x - K)^2$. We treat each k :

- $\tilde{Y}_0: y^3 = x^2(x - K)^2 = [x(x - K)]^2$. The substitution $v^3 = x(x - K)$ identifies $\tilde{Y}_0$ with $Y_1: v^3 = x(x - K)$, the cube-root elliptic curve of Theorem 4.3, and $1/y^2 = 1/v^4 = v^{-1}/v^3 = v^{-1}/(x(x - K))$. After this identification $\tilde{J}_0$ is a non-zero rational multiple of an Abelian integral on Y_1 , hence non-elementary by Theorem 4.3(ii).
- $\tilde{Y}_1: y^3 = x(x - K)^2$. The rational parametrization $x = Ks^3/(s^3 - 1)$, $y = Ks/(s^3 - 1)$ has Jacobian $dx = -3Ks^2 ds/(s^3 - 1)^2$ and gives $x^{1/3}(x - K)^{2/3} = Ks/(s^3 - 1)$ (with consistent cube-root branches), whence

$$\tilde{J}_1 = - \int \frac{\tilde{\varphi}_1\left(\frac{Ks^3}{s^3-1}\right) s ds}{s^3 - 1}.$$

- $\tilde{Y}_2: y^3 = (x - K)^2$. The substitution $u = (x - K)^{1/3}$ gives $x = u^3 + K$, $dx = 3u^2 du$, $(x - K)^{2/3} = u^2$, whence

$$\tilde{J}_2 = \int \tilde{\varphi}_2(u^3 + K) du.$$

The reductions in (15) follow, and elementarity is equivalent to $\tilde{H}_0 \equiv 0$ by the same Liouville-theoretic argument as in Theorem 4.3(ii) (any two of $\tilde{J}_0, \tilde{J}_1, \tilde{J}_2$ being elementary while the third is not would violate Theorem 2.4).

(ii) The decomposition $G = G_0 + G_1y + G_2y^2$ is the unique representation of $\mathbb{C}(t, y)$ as a free $\mathbb{C}(t)$ -module on the basis $\{1, y, y^2\}$. Multiplying through by y/y in the second and third terms,

$$G_1 y = \frac{G_1 y \cdot y^2}{y^2} = \frac{G_1 R}{y^2} = \frac{G_1 R}{R^{2/3}}, \quad G_2 y^2 = \frac{G_2 y^2 \cdot y}{y} = \frac{G_2 R}{y} = \frac{G_2 R}{R^{1/3}},$$

which is (16). The three pieces are individually elementary or not by the rational case (trivially), Theorem 4.3, and part (i) respectively, and their sum is elementary if and only if each piece is, again by Liouville's theorem. $\square$

Remark 4.10 (A duality between the two exponents). Comparing (11) and (14), the obstructive eigenspace shifts from V_1 at exponent $1/3$ to V_0 at exponent $2/3$. Geometrically, this reflects which 1-form of $\mathbb{C}(t, y)$ is of the first kind: on the cube-root elliptic curve $y^3 = x(x - K)$, the differential dx/y^2 is the unique (up to scale) holomorphic form, while dx/y has a pole at infinity. The eigenspace assignment of the obstruction tracks precisely this dichotomy.

Remark 4.11 (Theorem 2 analog). For R cubic, the full symmetry group of the unordered triple $\{a, b, c\}$ is the symmetric group S_3 , comprising the cyclic $\mathbb{Z}/3$ subgroup considered above together with three transpositions (Möbius involutions swapping pairs). Combining the cyclic and transposition characters yields a six-term sum criterion analogous to the Klein four-group condition (7) of Theorem 3.5; we omit the details.

5 Worked examples

We now illustrate the framework with a sequence of explicit calculations. Throughout, $\omega = e^{2\pi i/3}$.

5.1 Square-root: a basic Goursat example

Example 5.1. Take $R(t) = (t^2 - 1)(t^2 - 4)$ and $F(t) = t$. The roots are $\{\pm 1, \pm 2\}$, and the three Möbius involutions of Lemma 3.2 are computed directly:

$$S_1(t) = -t, \quad S_2(t) = 2/t, \quad S_3(t) = -2/t.$$

Verification of the V_4 -sum:

$$F + F \circ S_1 + F \circ S_2 + F \circ S_3 = t + (-t) + \frac{2}{t} + \left(-\frac{2}{t}\right) = 0.$$

The character decomposition (with the labelling of Theorem 3.5's proof, $\chi_j(S_j) = +1$) gives

$$F^{(1)} = 0, \quad F^{(2)} = \frac{t^2 + 2}{2t}, \quad F^{(3)} = \frac{t^2 - 2}{2t}.$$

(One checks directly: $F^{(2)}(2/t) = F^{(2)}(t)$ and $F^{(3)}(-2/t) = F^{(3)}(t)$.) Apply Theorem 3.3 to $F^{(3)}$ with the involution $S_1(t) = -t$ (fixed points $\alpha = 0, \beta = \infty$): since $\beta = \infty$ the substitution is just $u = t$, and (5) becomes the trivial $R(u) = (u^2 - 1)(u^2 - 4)$, already even. Apply $x = u^2$ to obtain

$$\int F^{(3)} \frac{dt}{\sqrt{R}} = \int \frac{t^2 - 2}{2t} \frac{dt}{\sqrt{R(t)}} = \frac{1}{2} \int \frac{(x - 2) dx}{2x \sqrt{(x - 1)(x - 4)}}.$$

A similar computation handles $F^{(2)}$. Adding the two contributions and simplifying yields

$$\int \frac{t dt}{\sqrt{(t^2 - 1)(t^2 - 4)}} = \frac{1}{2} \log(2t^2 - 5 + 2\sqrt{(t^2 - 1)(t^2 - 4)}) + C,$$

which one verifies by direct differentiation. □

5.2 Cube root: the canonical setting $R(t) = t^3 - 1$

For the next three examples, $R(t) = t^3 - 1$. The roots are $\{1, \omega, \omega^2\}$, and the order-3 Möbius transformation cyclically permuting them is simply $S(t) = \omega t$. The fixed points of S are $\alpha = 0, \beta = \infty$, so the substitution $z = (t - \alpha)/(t - \beta)$ degenerates to $z = t$ and no preliminary change of coordinates is required. The constants in (9) are $c = 1, K = 1$, so the canonical radical is $\sqrt[3]{z^3 - 1} = \sqrt[3]{t^3 - 1}$.

In this setting, the eigendecomposition of $H(z) = F(z)$ under $z \mapsto \omega z$ is exactly the eigendecomposition of F under multiplication by ω .

Example 5.2 ($F = 1$: elementary). $H(z) = 1 \in V_0$, so $H_1 \equiv 0$ and Theorem 4.3 predicts elementarity. With $\varphi_0(x) = 1$ and $K = 1$:

$$\int \frac{dt}{\sqrt[3]{t^3 - 1}} = J_0 = \frac{1}{3} \int \frac{dx}{\sqrt[3]{x^2(x - 1)}}.$$

Apply the parametrization (13) of Y_0 : $y^3 = x^2(x - 1)$:

$$x = \frac{1}{1 - w^3}, \quad y = \frac{w}{1 - w^3}, \quad \frac{dx}{y} = \frac{3w}{1 - w^3} dw.$$

Hence

$$J_0 = \frac{1}{3} \int \frac{dx}{y} = \int \frac{w \, dw}{1 - w^3}.$$

The remaining integrand has the partial-fraction decomposition

$$\frac{w}{1 - w^3} = \frac{w}{(1 - w)(1 + w + w^2)} = \frac{A}{1 - w} + \frac{Bw + C}{1 + w + w^2}$$

with $A = \frac{1}{3}$, $B = \frac{1}{3}$, $C = -\frac{1}{3}$, giving

$$\int \frac{w \, dw}{1 - w^3} = -\frac{1}{3} \log(1 - w) + \frac{1}{6} \log(1 + w + w^2) - \frac{1}{\sqrt{3}} \arctan\left(\frac{2w + 1}{\sqrt{3}}\right) + C.$$

Inverting the parametrization to recover w in terms of t : from $x = t^3 = 1/(1 - w^3)$ we get $1 - w^3 = 1/t^3$, hence $w^3 = (t^3 - 1)/t^3$ and

$$w = \frac{\sqrt[3]{t^3 - 1}}{t}.$$

Substituting back yields the explicit antiderivative:

$$\begin{aligned} \int \frac{dt}{\sqrt[3]{t^3 - 1}} &= -\frac{1}{3} \log\left(1 - \frac{\sqrt[3]{t^3 - 1}}{t}\right) + \frac{1}{6} \log\left(1 + \frac{\sqrt[3]{t^3 - 1}}{t} + \frac{\sqrt[3]{(t^3 - 1)^2}}{t^2}\right) \\ &\quad - \frac{1}{\sqrt{3}} \arctan\left(\frac{1}{\sqrt{3}}\left(2\frac{\sqrt[3]{t^3 - 1}}{t} + 1\right)\right) + C. \end{aligned}$$

One verifies this directly: with $w(t) = \sqrt[3]{t^3 - 1}/t$, a short calculation gives $dw/dt = 1/(t^2(t^3 - 1)^{2/3})$, while $w/(1 - w^3) = w \cdot t^3 = t^2 \sqrt[3]{t^3 - 1}$, so that the chain rule yields $d/dt \int w \, dw/(1 - w^3) = w/(1 - w^3) \cdot dw/dt = 1/\sqrt[3]{t^3 - 1}$, as required. $\square$

Example 5.3 ($F = t^2$: trivially elementary). $H(z) = z^2 \in V_2$, so $H_0 = H_1 = 0$ and only J_2 contributes. With $\varphi_2(x) = 1$:

$$J_2 = \frac{1}{3} \int \frac{dx}{(x - 1)^{1/3}} \Big|_{x=t^3} = \frac{1}{3} \cdot \frac{3}{2} (x - 1)^{2/3} \Big|_{x=t^3} + C = \frac{1}{2} (t^3 - 1)^{2/3} + C.$$

This recovers the obvious identity $\int t^2 \, dt/\sqrt[3]{t^3 - 1} = \frac{1}{2} (\sqrt[3]{t^3 - 1})^2 + C$, which one would obtain by inspection (the integrand is, up to a factor of 3, the derivative of $(t^3 - 1)^{2/3}$). The eigenvalue framework is consistent with — and gives a different proof of — this elementary observation. $\square$

Example 5.4 ($F = t$: not elementary). $H(z) = z \in V_1$, so $H_0 = H_2 = 0$ and only J_1 contributes. With $\varphi_1(x) = 1$ and $K = 1$:

$$\int \frac{t \, dt}{\sqrt[3]{t^3 - 1}} = J_1 = \frac{1}{3} \int \frac{dx}{\sqrt[3]{x(x - 1)}}.$$

The right-hand integral is the canonical period integral of the genus-1 curve $y^3 = x(x - 1)$, which is a non-elementary Abelian integral by the discussion in Theorem 4.3(ii).

Concretely, this is a cube-root analog of an incomplete elliptic integral; the corresponding complete period of $y^3 = x(x - 1)$ belongs to the family of CM-type periods controlled by the Chowla–Selberg formula [5, 9]. $\square$

The contrast among the three examples 5.2–5.4 is striking: $\int dt/\sqrt[3]{t^3 - 1}$ is elementary (yet non-trivially so), $\int t^2 \, dt/\sqrt[3]{t^3 - 1}$ is elementary (and trivially), and $\int t \, dt/\sqrt[3]{t^3 - 1}$ is genuinely transcendental — a definite “elliptic-like” integral living on $y^3 = x(x - 1)$.

5.3 A combined integrand

Example 5.5. Take $R(t) = t^3 - 1$ and $F(t) = 1 + 5t^2 - 7t/(t^3 + 1)$. Decompose F :

$$F = 1 + 5t^2 + \tilde{F}(t), \quad \tilde{F}(t) := -\frac{7t}{t^3+1}.$$

The first two terms are in V_0 and V_2 respectively. The third term $\tilde{F}(t)$ is of the form $t \cdot \varphi(t^3)$ with $\varphi(x) = -7/(x+1)$, hence in V_1 . By Theorem 4.3, the integral

$$\int F \frac{dt}{\sqrt[3]{t^3-1}}$$

is elementary if and only if $\tilde{F} \equiv 0$, which it is not. Hence the integral splits as

$$\underbrace{\int \frac{dt}{\sqrt[3]{t^3-1}}}_{\text{Example 5.2}} + 5 \underbrace{\int \frac{t^2 dt}{\sqrt[3]{t^3-1}}}_{\text{Example 5.3}} - \frac{7}{3} \int \frac{dx}{(x+1)\sqrt[3]{x(x-1)}},$$

where the last integral is a non-elementary Abelian integral on $Y_1: y^3 = x(x-1)$. The framework therefore not only detects elementarity but also *quantifies the obstruction* as a specific Abelian integral on a known curve. $\square$

5.4 A non-canonical S : shifted cube

Example 5.6. Consider $R(t) = (t-1)(t-2)(t-3)$ and the order-3 Möbius transformation S with $S(1) = 2$, $S(2) = 3$, $S(3) = 1$. Solving the three linear conditions gives

$$S(t) = \frac{5t-13}{3t-7}, \quad \alpha, \beta = 2 \pm \frac{i}{\sqrt{3}}.$$

A direct check confirms $S(1) = -8/-4 = 2$, $S(2) = -3/-1 = 3$, $S(3) = 2/2 = 1$, and the fixed points are the roots of $3t^2 - 12t + 13 = 0$.

The substitution $z = (t - \alpha)/(t - \beta)$ brings S into canonical form $z \mapsto \omega z$ and produces, by Lemma 4.2, a polynomial $c(z^3 - K)$ for explicit c and K . One can then test any rational F by computing its eigencomponents in z -coordinates. For F chosen to lie entirely in V_0 or V_2 in the new coordinates, one obtains explicit elementary antiderivatives via the procedure of Theorem 4.3; for F with non-vanishing V_1 -component, the integral involves a non-elementary period of the genus-1 curve $y^3 = x(x-K)$. $\square$

5.5 A degenerate cubic: degree-2 radicand

Example 5.7. We illustrate Corollary 4.7 concretely with $R(t) = t^2 - 1$, viewed as a cubic with one root at ∞ (cf. Remark 4.6). Here $r_1 = 1$, $r_2 = -1$, so the formula of Remark 4.6(b) gives

$$S(t) = \frac{1 \cdot t - (1+1+1)}{t - (-1)} = \frac{t-3}{t+1},$$

with fixed-point equation $t^2 + 3 = 0$ and so $\alpha = i\sqrt{3}$, $\beta = -i\sqrt{3}$. A direct check confirms $S(1) = -1$, $S(-1) = \infty$, $S(\infty) = 1$.

Substituting $z = (t - \alpha)/(t - \beta)$ with inverse $t(z) = i\sqrt{3}(1+z)/(1-z)$:

$$R(t(z))(1-z)^3 = [t(z)^2 - 1](1-z)^3 = 4(z^3 - 1),$$

confirming $K = 1$ as predicted by Remark 4.6(a), with $c = 4$. The function (11) for $F = 1$ is

$$H(z) = \frac{2i\sqrt{3}}{\sqrt[3]{4}(1-z)} = \frac{c_0}{1-z}, \quad c_0 := \frac{2i\sqrt{3}}{\sqrt[3]{4}} \neq 0.$$

By Corollary 4.7,

$$\int \frac{dt}{\sqrt[3]{t^2 - 1}} \text{ is not elementary.}$$

By Theorem 4.3(ii), the obstruction takes the explicit form

$$J_1 = \frac{1}{3} \int \frac{\varphi_1(x) dx}{\sqrt[3]{x(x-1)}}, \quad \varphi_1(x) = \frac{c_0}{1-x},$$

a non-elementary Abelian integral on the cube-root elliptic curve $y^3 = x(x-1)$. The integrals J_0 and J_2 contribute elementary pieces (with arctan and log terms via the parametrization (13) and the trivial reduction for J_2), but they cannot cancel the transcendental J_1 .

The contrast with the square-root case $\int dt/\sqrt{t^2-1} = \operatorname{arccosh}(t) + C$ is geometric: the curve $y^2 = t^2 - 1$ has genus 0 (it is a smooth conic), but $y^3 = t^2 - 1$ has genus 1. $\square$

6 Concluding remarks

Higher radicals.

Theorem 4.3 extends to n -th roots, considering integrands $\int F(t) dt / \sqrt[n]{R(t)}$ together with order- n Möbius transformations S cyclically permuting roots of R . After the substitutions $z = (t - \alpha)/(t - \beta)$ and $x = z^n$, the differential decomposes into n eigencomponents under $z \mapsto \zeta z$ (with $\zeta = e^{2\pi i/n}$), the k -th of which lands on the curve

$$Y_k: y^n = x^{n-1-k}(x - K), \quad k = 0, 1, \dots, n-1.$$

A direct application of Riemann–Hurwitz to the projection $Y_k \rightarrow \mathbb{P}^1$, $(x, y) \mapsto x$, gives

$$g(Y_k) = \frac{n+1 - \gcd(n, k) - \gcd(n, k+1)}{2}. \quad (17)$$

Since $\gcd(n, k)$ and $\gcd(n, k+1)$ are coprime divisors of n (as k and $k+1$ are coprime), their sum equals $n+1$ if and only if one of them equals n , which occurs precisely for $k = 0$ or $k = n-1$. Therefore

$$g(Y_k) = 0 \iff k \in \{0, n-1\},$$

and every interior $k \in \{1, \dots, n-2\}$ gives a curve of strictly positive genus. The two outermost eigencomponents always descend to elementary integrals — via the rational parametrizations $x = K/(1-w^n)$ and $x = K+w^n$ for Y_0 and Y_{n-1} respectively — while the interior components are generically transcendental.

For small n :

- $n = 2$ (*no interior k*). Both eigencomponents descend to genus-0 curves; recovering Goursat's theorem (Theorem 3.3).
- $n = 3$ (*one interior $k = 1$*). The unique obstruction is $Y_1: y^3 = x(x-K)$ of genus 1, the cube-root analog of a Weierstrass elliptic curve — this is Theorem 4.3.
- $n = 4$ (*two interior $k \in \{1, 2\}$, both genus 1*). A particularly clean rigidity emerges. For $k = 1$, the substitution $u = y^2/x$ on $Y_1: y^4 = x^2(x-K)$ gives $u^2 = x-K$, hence $x = u^2 + K$ and

$$y^2 = u \cdot x = u(u^2 + K) = u^3 + Ku,$$

the *lemniscate elliptic curve* with j -invariant 1728. For $k = 2$, $Y_2: y^4 = x(x-K)$ carries the order-4 automorphism $y \mapsto iy$, which descends to an order-4 automorphism of its elliptic quotient and therefore forces $j = 1728$. The two obstructions thus both reduce, over $\mathbb{C}$, to

the unique elliptic curve with complex multiplication by $\mathbb{Z}[i]$ — the rigidity is forced by the underlying $\mathbb{Z}/4$ -symmetry of the cover.

- $n = 5$ (three interior $k \in \{1, 2, 3\}$, all genus 2). By (17) all three interior curves Y_1, Y_2, Y_3 have genus 2, giving three independent transcendental obstructions on hyperelliptic curves.
- $n \geq 6$. The genera of the Y_k vary non-monotonically with k and the obstruction curves are no longer pairwise isomorphic. For example at $n = 6$ the interior values $k \in \{1, 4\}$ give genus 2 and $k \in \{2, 3\}$ give genus 1 (cube-root and square-root elliptic respectively), a mixed picture lacking the rigidity present at $n \in \{3, 4\}$.

Theorem 2 analog for $n = 3$.

For R cubic, the full automorphism group of the unordered triple of roots is S_3 , with character table comprising the trivial, sign, and 2-dimensional irreducible representations. A complete cube-root analog of Theorem 3.5 should sum over all of S_3 (including transpositions), and the obstruction — which corresponds in the cyclic case to a single eigenspace — now corresponds to the 2-dimensional irreducible component. Working this out and matching to the cube-root analog of the Klein four-group decomposition is straightforward but notationally heavier.

Algorithmic perspective.

The Risch–Trager–Bronstein algorithm [12, 13, 7, 16, 3, 4] provides, in principle, a uniform decision procedure for elementary integration over algebraic function fields, including cube-radical integrands; in this sense it subsumes both Goursat’s theorem and our cube-root analog. In practice, however, the algebraic case of the Risch algorithm is extremely complex and has never been implemented in fully general form in any computer algebra system (see also the discussion in [2]); even the FriCAS implementation [16], which remains the most complete, is known to be unexpectedly slow on inputs of modest size and to fail outright on a variety of integrands handled effortlessly by classical methods — including a number of Goursat-type pseudo-elliptics noted in the recent literature and on `sci.math.symbolic` [18]. Moreover, the Risch–Trager–Bronstein output, when produced, is usually opaque: it expresses an antiderivative in a specific algebraic form without revealing the underlying geometric structure. The reductions of Theorems 3.3 and 4.3 are valuable precisely because they preserve and explain the structure, run in polynomial time in the degrees of R and F , and often produce strictly simpler closed forms than the algorithmic output.

Differential Galois theory.

At a higher level, the question of which integrals are elementary is governed by the differential Galois group of the relevant Picard–Vessiot extension; we refer to the standard texts of Magid [11], van der Put–Singer [17], and Crespo–Hajto [6] for the algebraic theory, and to Khovanskii [10] for a topological viewpoint that includes Liouville’s results on solvability in finite terms among its main applications. For curves of positive genus, the Galois-theoretic obstructions to elementarity are governed by the Albanese variety and the structure of the divisor class group [1]. Goursat’s theorem and its cube-root analog can be viewed as explicit calculations of these obstructions in the presence of additional curve automorphisms, and as concrete instances of the general principle that the more symmetry a curve has, the more likely a given Abelian integral is to degenerate to elementary functions.

A Mathematica implementation

We provide a self-contained *Mathematica* package implementing the diagnostic algorithms of Theorems 3.5, 4.3, and Corollary 4.9. The entry point is

$$\text{IntegrateGoursat}[F, R, t, p], \quad p \in \{\frac{1}{2}, \frac{1}{3}, \frac{2}{3}\},$$

which takes a rational integrand $F \in \mathbb{C}(t)$, a polynomial $R \in \mathbb{C}[t]$ with simple roots, and an exponent p specifying that the integrand is $F(t)/R(t)^p$. The package computes the relevant character or eigenspace projections and returns a Mathematica association keyed by "status". Two outcomes are possible:

- "status" $\rightarrow$ "elementary": the projection criterion of the relevant theorem is satisfied. The returned association contains the explicit reduction to a rational integrand — via Theorem 3.3 for $p = \frac{1}{2}$ (yielding $\int G(x) dx / \sqrt{Q(x)}$ with Q quadratic), via the parametrizations of Y_0 and Y_2 for $p = \frac{1}{3}$ (yielding rational integrands in w and u), and via the parametrizations of $\tilde{Y}_1$ and $\tilde{Y}_2$ for $p = \frac{2}{3}$ (yielding rational integrands in s and u).
- "status" $\rightarrow$ "obstructed": the projection criterion fails, and the obstruction — a non-zero rational function in the V-isotypic subspace identified by the relevant theorem — is returned. Specifically, $F^{(0)}$ for $p = \frac{1}{2}$, H_1 for $p = \frac{1}{3}$, or $\tilde{H}_0$ for $p = \frac{2}{3}$. By Theorems 3.5, 4.3(ii), and Corollary 4.9(ii), this verdict implies non-elementarity *whenever the descended differential on the appropriate obstruction curve is of the second kind* — a sufficient condition that holds in particular when F is a polynomial in t , and more generally when the rational function φ_k (or $\tilde{\varphi}_k$) extracted from the obstruction has poles only at branch points of the cube-root cover. For F with poles at non-branch points of the cube-root cover, the obstructed status is necessary for non-elementarity but not sufficient: a complete Risch–Trager–Bronstein analysis [16, 4] would be required to decide elementarity definitively. In practice, for the polynomial- F inputs that motivate the development, the obstructed verdict is a complete certificate of non-elementarity.

Crucially, the diagnostic phase never invokes Mathematica's **Integrate**: every step is the constructive procedure from the proofs of Theorems 3.3–4.3 and Corollary 4.9, performed symbolically in rational arithmetic. The auto-extracting two-argument form **IntegrateGoursat**[**integrand**, **t**] (described below) does call **Integrate** recursively, but only on the rational reductions produced when the verdict is "elementary".

The code mirrors the proofs directly. The square-root branch (**GoursatTest**) computes the three Möbius involutions of Lemma 3.2 via the explicit formula (3), then the four V_4 -character projections from the proof of Theorem 3.5; for each non-zero non-trivial projection $F^{(j)}$ ($j \in \{1, 2, 3\}$) it picks an involution S_k ($k \neq j$) under which $F^{(j)}$ is anti-invariant (preferring one whose fixed-point pair contains ∞ , which gives the simplest substitution) and applies the constructive proof of Theorem 3.3 to produce the reduction $((\alpha - \beta)/(2\sqrt{c})) \int G(u) du / \sqrt{Q(u)}$ explicitly, where u is a substitution variable supplied by the caller. As noted in the proof of Theorem 3.5, the cubic case is handled uniformly by treating ∞ as the fourth ramification point: when **Solve**[**R == 0**, **t**] returns three finite roots, the package appends **Infinity** so that the V_4 of involutions is constructed as in the quartic case. The implementation of **MobiusInvolution** dispatches to a closed-form expression $S(t) = (ct + ab - c(a + b))/(t - c)$ (the limit of the general formula (3) as one of the four arguments tends to infinity) when called with **Infinity** in any slot, and applies **canonic** — a thin wrapper for **Cancel**[**Together**[..., **Extension** $\rightarrow$ **Automatic**], **Extension** $\rightarrow$ **Automatic**] — throughout to carry out simplifications across the algebraic-number extensions generated by the roots of R .

The package exposes two entry points. The five-argument form **IntegrateGoursat**[**F**, **R**, **t**, **p**, **u**] (with p defaulting to $1/2$) returns the diagnostic Association described above, leaving the rational reductions in u for the caller to inspect or integrate. The two-argument form

`IntegrateGoursat[integrand, t]` parses an integrand of the form $F(t) R(t)^{-p}$ via `ParseGoursatIntegrand`, dispatches to the appropriate test, and — when the verdict is "elementary" — recursively invokes Mathematica's `Integrate` on each rational reduction and back-substitutes through the explicit substitution chain (recorded in the "back" or "J0back"/"J2back" fields of the diagnostic output) to produce a closed-form antiderivative in t . The back-substitution rules follow from the canonical-form lemmas: for $p = 1/2$ the chain is $u_{\text{out}} = ((t - \alpha)/(t - \beta))^2$; for $p = 1/3$, $u_{\text{out}}^{J_0} = R(t)^{1/3}(\alpha - \beta)/(c^{1/3}(t - \alpha))$ and $u_{\text{out}}^{J_2} = R(t)^{1/3}(\alpha - \beta)/(c^{1/3}(t - \beta))$, with the corresponding $\beta = \infty$ specialisations; the $p = 2/3$ case is dual via $u_{\text{out}}^{J_1} = c^{1/3}(t - \alpha)/(R(t)^{1/3}(\alpha - \beta))$, the reciprocal of the J_0 rule. Every back-substitution is the composition of (i) the Möbius substitution $z = (t - \alpha)/(t - \beta)$, (ii) the eigenspace coordinate $x = z^3$ (or $x = z^2$ for $p = 1/2$), and (iii) the rational parametrisation of the relevant genus-zero curve Y_k .

The cube-root branches share their setup — the order-3 Möbius transformation of Lemma 4.1, the change of variable $z = (t - \alpha)/(t - \beta)$, and the projection formula (12) — and differ only in which function is decomposed and which eigenspace obstructs. For exponent $1/3$ (`CubicTest`), the relevant function is H from (11), the obstruction lives in V_1 , and the reductions land on Y_0 and Y_2 via the parametrizations $x = K/(1 - w^3)$ and $u = \sqrt[3]{x - K}$. For exponent $2/3$ (`CubicTest23`), the function is $\tilde{H}$ from (14), the obstruction lives in V_0 , and the reductions land on $\tilde{Y}_1$ and $\tilde{Y}_2$ via the parametrizations $x = Ks^3/(s^3 - 1)$ and $u = \sqrt[3]{x - K}$. The case $\beta = \infty$ — which arises whenever 0 or ∞ is fixed by S , e.g. for $R(t) = t^3 - 1$ with $S(t) = \omega t$ — is handled in both branches by the explicit closed form for the cyclic Möbius transformation derived in Remark 4.6.

Source code

```
(* === GoursatAppendix.m ===
Diagnostic for pseudo-elementarity of integrals
  Integrate[F[t]/R[t]^p, t]
for p in {1/2, 1/3, 2/3}. Reductions are constructive (Theorems
3.3, 3.5, 4.3, Cor. 4.8); the auto-extracting entry point
IntegrateGoursat[integrand, t] additionally calls Mathematica's
Integrate on the rational reductions to produce a closed-form
antiderivative in t.
*)

(* === Helper: canonicalise across algebraic-number extensions === *)
canonic[e_] :=
  Cancel[Together[e, Extension -> Automatic], Extension -> Automatic];

(* === Square-root case (Theorems 3.3 and 3.5) === *)

(* Möbius involution swapping pairs {a,b} and {c,d}. Handles Infinity *)
(* in any slot by reducing to the canonical case d -> oo. *)
MöbiusInvolution[{a_, b_}, {c_, d_}, t_] :=
  Which[
    a === Infinity, MöbiusInvolution[{c, d}, {b, a}, t],
    b === Infinity, MöbiusInvolution[{c, d}, {a, b}, t],
    c === Infinity, MöbiusInvolution[{a, b}, {d, c}, t],
    d === Infinity, canonic[(c t + a b - c (a + b))/(t - c)],
    True, canonic[((a b - c d) t + (a + b) c d - (c + d) a b)/
      (((a + b) - (c + d)) t - (a b - c d))]
  ];

V4Projections[F_, {S1_, S2_, S3_}, t_] :=
  Module[{f, f1, f2, f3},
    {f, f1, f2, f3} = {F, F /. t -> S1, F /. t -> S2, F /. t -> S3};
    canonic /@ {(f + f1 + f2 + f3)/4, (f + f1 - f2 - f3)/4,
      (f - f1 + f2 - f3)/4, (f - f1 - f2 + f3)/4};
```

```

ToFunctionOfSquare[H_, u_, x_] :=
  If[PossibleZeroQ[Together[H]], 0,
    Module[{num, den, dN, dD},
      {num, den} = Through[{Numerator, Denominator}[canonic[H]]];
      dN = Exponent[num, u]; dD = Exponent[den, u];
      Sum[Coefficient[num, u, 2 k] x^k, {k, 0, dN/2}]/
      Sum[Coefficient[den, u, 2 k] x^k, {k, 0, dD/2}]]];

(* Reduction: takes uOut as the user-facing output variable; *)
(* the intermediate Mobius substitution variable is local. *)
GoursatReduction[Fj_, R_, t_, S_, uOut_] :=
  Module[{u, fps, alpha, beta, tu, Rfact, lc, Q, Fu, gu, gx, pre, back},
    fps = t /. Solve[S == t, t];
    {alpha, beta} = If[Length[fps] == 1, {fps[[1]], Infinity}, fps];
    tu = If[beta == Infinity, alpha + u, (alpha - beta u)/(1 - u)];
    Rfact = If[beta == Infinity, Expand[R /. t -> tu],
      Expand[canonic[(R /. t -> tu) (1 - u)^4]]];
    lc = Coefficient[Rfact, u, 4];
    Q = Sum[Coefficient[Rfact, u, 2 k]/lc uOut^k, {k, 0, 2}];
    Fu = canonic[Fj /. t -> tu]; gu = canonic[Fu/u];
    gx = ToFunctionOfSquare[gu, u, uOut];
    pre = If[beta == Infinity, 1, alpha - beta]/(2 Sqrt[lc]);
    back = If[beta == Infinity, uOut -> (t - alpha)^2,
      uOut -> ((t - alpha)/(t - beta))^2];
    <|"S" -> S, "alpha" -> alpha, "beta" -> beta,
      "prefactor" -> pre, "G" -> gx, "Q" -> Q, "back" -> back|>;

PickAntiInvolution[j_, S_, t_] :=
  Module[{cands, withInf},
    cands = Complement[{1, 2, 3}, {j}];
    withInf = Select[cands,
      Length[t /. Solve[S[[#]] == t, t]] == 1 &];
    S[[If[Length[withInf] > 0, First[withInf], First[cands]]]]];

GoursatTest[F_, R_, t_, uOut_] :=
  Module[{roots, S, p, reds},
    roots = DeleteDuplicates[t /. Solve[R == 0, t]];
    (* The cubic case is identical with r_4 = Infinity. *)
    If[Length[roots] == 3, AppendTo[roots, Infinity]];
    If[Length[roots] != 4, Return[$Failed]];
    S = With[{r = roots},
      {MobiusInvolution[r[[{1, 2}]], r[[{3, 4}]], t],
        MobiusInvolution[r[[{1, 3}]], r[[{2, 4}]], t],
        MobiusInvolution[r[[{1, 4}]], r[[{2, 3}]], t]};
    p = V4Projections[F, S, t];
    If[!PossibleZeroQ[p[[1]]],
      Return[<|"status" -> "obstructed",
        "involutions" -> S, "obstruction" -> p[[1]]|>]];
    reds = Association @ Table[
      If[!PossibleZeroQ[p[[j + 1]]],
        j -> GoursatReduction[p[[j + 1]], R, t,
          PickAntiInvolution[j, S, t], uOut],
        Nothing], {j, 1, 3}];
    <|"status" -> "elementary", "involutions" -> S,
      "F0" -> p[[1]], "F1" -> p[[2]],
      "F2" -> p[[3]], "F3" -> p[[4]],
      "reductions" -> reds|>;

(* == Cube-root case (Theorem 4.3, Corollary 4.8) == *)

CyclicMobius[roots_List, t_] :=
  Module[{r, pos, A, B, C, sol},
    r = roots;

```

```

pos = FirstPosition[r, Infinity, {0}, 1][[1]];
If[pos > 0,
  r = RotateLeft[r, pos];
  Together[(r[[1]] t -
    (r[[1]]^2 - r[[1]] r[[2]] + r[[2]]^2))/(t - r[[2]]),
  sol = First[Solve[
    {(A r[[1]] + B)/(C r[[1]] + 1) == r[[2]],
    (A r[[2]] + B)/(C r[[2]] + 1) == r[[3]],
    (A r[[3]] + B)/(C r[[3]] + 1) == r[[1]]}, {A, B, C}]];
  canonic[(A t + B)/(C t + 1) /. sol]]];

EigenProjection[H_, z_, k_Integer] :=
  With[{w = Exp[2 Pi I/3]},
    canonic[Simplify[(H + w^(-k) (H /. z -> w z) +
      w^(-2 k) (H /. z -> w^2 z))/3]]];

ToFunctionOfCube[H_, z_, x_] :=
  If[PossibleZeroQ[Together[H]], 0,
    Module[{num, den, dN, dD},
      {num, den} = Through[{Numerator, Denominator}[canonic[H]]];
      dN = Exponent[num, z]; dD = Exponent[den, z];
      Sum[Coefficient[num, z, 3 k] x^k, {k, 0, dN/3}]/
      Sum[Coefficient[den, z, 3 k] x^k, {k, 0, dD/3}]]];

CubicTest[F_, R_, t_, uOut_] :=
  Module[{roots, S, fps, alpha, beta, z, x, tz, Rz, cval, K,
    H, H0, H1, H2, phi0, psi2, J0, J2, J0back, J2back},
    roots = DeleteDuplicates[t /. Solve[R == 0, t]];
    If[Length[roots] == 2, AppendTo[roots, Infinity]];
    If[Length[roots] != 3, Return[$Failed]];
    S = CyclicMobius[roots, t];
    fps = t /. Solve[S == t, t];
    {alpha, beta} = If[Length[fps] == 1, {fps[[1]], Infinity}, fps];
    tz = If[beta === Infinity, alpha + z, (alpha - beta z)/(1 - z)];
    Rz = If[beta === Infinity, canonic[R /. t -> tz],
      canonic[(R /. t -> tz) (1 - z)^3]];
    cval = Coefficient[Expand[Rz], z, 3];
    K = -Coefficient[Expand[Rz], z, 0]/cval;
    H = Together[If[beta === Infinity,
      (F /. t -> tz)/cval^(1/3),
      (alpha - beta) (F /. t -> tz)/(cval^(1/3) (1 - z))]];
    {H0, H1, H2} = EigenProjection[H, z, #] & /@ {0, 1, 2};
    If[!PossibleZeroQ[Simplify[H1]],
      Return[<|"status" -> "obstructed", "S" -> S,
        "alpha" -> alpha, "beta" -> beta, "K" -> K,
        "obstruction" -> H1|>]];
    phi0 = ToFunctionOfCube[H0, z, x];
    psi2 = ToFunctionOfCube[Together[H2/z^2], z, x];
    J0 = Together[(phi0 /. x -> K/(1 - uOut^3)) uOut/(1 - uOut^3)];
    J2 = Together[(psi2 /. x -> uOut^3 + K) uOut];
    J0back = If[beta === Infinity,
      uOut -> R^(1/3)/(cval^(1/3) (t - alpha)),
      uOut -> R^(1/3) (alpha - beta)/(cval^(1/3) (t - alpha))];
    J2back = If[beta === Infinity,
      uOut -> R^(1/3)/cval^(1/3),
      uOut -> R^(1/3) (alpha - beta)/(cval^(1/3) (t - beta))];
    <|"status" -> "elementary", "S" -> S,
      "alpha" -> alpha, "beta" -> beta, "K" -> K, "c" -> cval,
      "H0" -> H0, "H2" -> H2,
      "J0integrand" -> J0, "J2integrand" -> J2,
      "J0back" -> J0back, "J2back" -> J2back|>];

CubicTest23[F_, R_, t_, uOut_] :=

```



```

Module[{roots, S, fps, alpha, beta, z, x, tz, Rz, cval, K,
  Htilde, H0, H1, H2, phi1, phi2, J1, J2, J1back, J2back},
  roots = DeleteDuplicates[t /. Solve[R == 0, t]];
  If[Length[roots] == 2, AppendTo[roots, Infinity]];
  If[Length[roots] != 3, Return[$Failed]];
  S = CyclicMobius[roots, t];
  fps = t /. Solve[S == t, t];
  {alpha, beta} = If[Length[fps] == 1, {fps[[1]], Infinity}, fps];
  tz = If[beta === Infinity, alpha + z, (alpha - beta z)/(1 - z)];
  Rz = If[beta === Infinity, canonic[R /. t -> tz],
    canonic[(R /. t -> tz) (1 - z)^3]];
  cval = Coefficient[Expand[Rz], z, 3];
  K = -Coefficient[Expand[Rz], z, 0]/cval;
  Htilde = Together[If[beta === Infinity,
    (F /. t -> tz)/cval^(2/3),
    (alpha - beta) (F /. t -> tz)/cval^(2/3)]];
  {H0, H1, H2} = EigenProjection[Htilde, z, #] & /@ {0, 1, 2};
  If[!PossibleZeroQ[Simplify[H0]],
    Return[<|"status" -> "obstructed", "S" -> S,
      "alpha" -> alpha, "beta" -> beta, "K" -> K,
      "obstruction" -> H0|>]];
  phi1 = ToFunctionOfCube[Together[H1/z], z, x];
  phi2 = ToFunctionOfCube[Together[H2/z^2], z, x];
  J1 = Together[-(phi1 /. x -> K uOut^3/(uOut^3 - 1)) uOut/(uOut^3 - 1)];
  J2 = Together[phi2 /. x -> uOut^3 + K];
  J1back = If[beta === Infinity,
    uOut -> cval^(1/3) (t - alpha)/R^(1/3),
    uOut -> cval^(1/3) (t - alpha)/(R^(1/3) (alpha - beta))];
  J2back = If[beta === Infinity,
    uOut -> R^(1/3)/cval^(1/3),
    uOut -> R^(1/3) (alpha - beta)/(cval^(1/3) (t - beta))];
  <|"status" -> "elementary", "S" -> S,
    "alpha" -> alpha, "beta" -> beta, "K" -> K, "c" -> cval,
    "H1" -> H1, "H2" -> H2,
    "J1integrand" -> J1, "J2integrand" -> J2,
    "J1back" -> J1back, "J2back" -> J2back|>];

(* === Unified entry point with substitution variable === *)

IntegrateGoursat[F_, R_, t_, p_ : 1/2, uOut_] :=
  Switch[p, 1/2, GoursatTest[F, R, t, uOut],
    1/3, CubicTest[F, R, t, uOut],
    2/3, CubicTest23[F, R, t, uOut],
    _, "Only p = 1/2, 1/3, 2/3 are supported"];

(* === Auto-extracting entry point: parses F, R, p from the      *)
(*   integrand and computes a closed-form antiderivative in t    *)
(*   by recursively invoking Mathematica's Integrate.            *)

ParseGoursatIntegrand[expr_, t_] :=
  Replace[expr, {
    f_. * Power[r_, q_]
      /; PolynomialQ[r, t] && MemberQ[{-1/2, -1/3, -2/3}, q] ->
        {canonic[f], r, -q},
    Power[r_, q_]
      /; PolynomialQ[r, t] && MemberQ[{-1/2, -1/3, -2/3}, q] ->
        {1, r, -q},
    _ -> $Failed]];

ComputeAntiderivative12[result_, t_, uOut_] :=
  Module[{total = 0},
    Do[With[{red = result["reductions"][j]},
      total = total + red["prefactor"] *

```

```

      (Integrate[red["G"]/Sqrt[red["Q"]], uOut] /. red["back"]]),
    {j, Keys[result["reductions"]]}];
total];

ComputeAntiderivative13[result_, t_, uOut_] :=
Module[{total = 0, J0 = result["J0integrand"], J2 = result["J2integrand"]},
  If[!PossibleZeroQ[J0],
    total = total + (Integrate[J0, uOut] /. result["J0back"])]];
  If[!PossibleZeroQ[J2],
    total = total + (Integrate[J2, uOut] /. result["J2back"])]];
total];

ComputeAntiderivative23[result_, t_, uOut_] :=
Module[{total = 0, J1 = result["J1integrand"], J2 = result["J2integrand"]},
  If[!PossibleZeroQ[J1],
    total = total + (Integrate[J1, uOut] /. result["J1back"])]];
  If[!PossibleZeroQ[J2],
    total = total + (Integrate[J2, uOut] /. result["J2back"])]];
total];

IntegrateGoursat[integrand_, t_] :=
Module[{parsed, F, R, p, uOut, result},
  parsed = ParseGoursatIntegrand[integrand, t];
  If[parsed === $Failed, Return[$Failed]];
  {F, R, p} = parsed;
  uOut = Unique["u"];
  result = IntegrateGoursat[F, R, t, p, uOut];
  If[!AssociationQ[result] || result["status"] != "elementary",
    Return[result]];
  Switch[p,
    1/2, ComputeAntiderivative12[result, t, uOut],
    1/3, ComputeAntiderivative13[result, t, uOut],
    2/3, ComputeAntiderivative23[result, t, uOut]]];

```

Sample sessions

We illustrate the package on four examples drawn from the body of the paper.

Example 5.1 (square-root, elementary). For $F = t$ and $R = (t^2 - 1)(t^2 - 4)$:

```
In[]:= IntegrateGoursat[t, (t^2 - 1)(t^2 - 4), t, 1/2, u]
```

returns "status" $\rightarrow$ "elementary" with $F^{(0)} = 0$ (verifying the V_4 -criterion), $F^{(1)} = 0$, and the two non-zero non-trivial projections $F^{(2)} = (t^2 + 2)/(2t)$ and $F^{(3)} = (t^2 - 2)/(2t)$. The "reductions" field contains, indexed by $j \in \{2, 3\}$, the data of Theorem 3.3 applied to $F^{(j)}$ with the involution $S = -t$ (preferred over the alternative anti-invariance involution because it has $\beta = \infty$, simplifying the substitution to $u = t$):

$$\text{prefactor} = \frac{1}{2}, \quad G_2(x) = \frac{x+2}{2x}, \quad G_3(x) = \frac{x-2}{2x}, \quad Q(x) = (x-1)(x-4).$$

Summing the two contributions and integrating the elementary integrals $\int G_j(x) dx/\sqrt{Q(x)}$ recovers the closed form derived by hand in Example 5.1.

Example 5.2 (cube-root, elementary). For $F = 1$, $R = t^3 - 1$:

```
In[]:= IntegrateGoursat[1, t^3 - 1, t, 1/3, u]
```

returns "status" $\rightarrow$ "elementary". The package detects $S(t) = \omega t$, $\alpha = 0$, $\beta = \infty$, $K = 1$, and $H(z) = 1 \in V_0$, so $H_0 = 1$, $H_2 = 0$. The rational reductions are

$$\text{J0integrand}(w) = \frac{w}{1 - w^3}, \quad \text{J2integrand}(u) = 0,$$

so $J_2 \equiv 0$ and $J_0 = \int w \, dw / (1 - w^3)$ is a genuinely rational integral, evaluable by partial fractions, recovering the closed form of Example 5.2.

Example 5.4 (cube-root, obstructed; non-elementary because F is polynomial). For $F = t$, $R = t^3 - 1$:

```
In[]:= IntegrateGoursat[t, t^3 - 1, t, 1/3, u]
```

returns "status" $\rightarrow$ "obstructed" with $K = 1$ and obstruction $H_1 = z$, exactly as predicted by Example 5.4.

Corollary 4.7 (cube-root, degree-2 radicand). For $F = 1$, $R = t^2 - 1$:

```
In[]:= IntegrateGoursat[1, t^2 - 1, t, 1/3, u]
```

returns "status" $\rightarrow$ "obstructed". Internally the package computes $S(t) = (t-3)/(t+1)$, fixed points $\alpha = i\sqrt{3}$, $\beta = -i\sqrt{3}$, the canonical-form constants $c = 4$, $K = 1$, and the obstruction $H_1 = c_0 z / (1 - z^3)$ with $c_0 = 2i\sqrt{3}/\sqrt[3]{4}$, matching the explicit eigendecomposition derived in the proof of Corollary 4.7 via the geometric-series identity $1/(1 - z) = (1 + z + z^2)/(1 - z^3)$.

Corollary 4.9 (the duality at exponent 2/3). The same input $F = 1$, $R = t^3 - 1$ that was elementary at exponent 1/3 is non-elementary at exponent 2/3:

```
In[]:= IntegrateGoursat[1, t^3 - 1, t, 2/3, u]
```

returns "status" $\rightarrow$ "obstructed" with obstruction $\tilde{H}_0 = 1$. Conversely, $F = t$ that was non-elementary at 1/3 is elementary at 2/3:

```
In[]:= IntegrateGoursat[t, t^3 - 1, t, 2/3, u]
```

returns "status" $\rightarrow$ "elementary" with $\tilde{H}_1 = z$, $\tilde{H}_2 = 0$, $\varphi_1 = 1$, and rational reductions

$$\text{J1integrand}(s) = -\frac{s}{s^3 - 1}, \quad \text{J2integrand}(u) = 0,$$

giving $\int t \, dt / (t^3 - 1)^{2/3} = -\int s \, ds / (s^3 - 1)$ via the parametrization $s = t / (t^3 - 1)^{1/3}$. This is the duality of Corollary 4.9: the obstructive eigenspace migrates from V_1 at exponent 1/3 to V_0 at exponent 2/3, exchanging the elementary and transcendental cases.

Auto-extracting form: closed-form antiderivative. The two-argument form parses the integrand and recursively invokes Mathematica's `Integrate` on the rational reductions:

```
In[]:= IntegrateGoursat[t^2/(t^3 - 1)^(1/3), t]
```

returns $\frac{1}{2}(t^3 - 1)^{2/3}$, the closed form of Example 5.3 obtained by integrating $\text{J2integrand}(u) = u$ and back-substituting $u \mapsto (t^3 - 1)^{1/3}$. When the diagnostic verdict is "obstructed", the auto-extracting form returns the diagnostic Association unchanged rather than attempting a closed form: e.g. `IntegrateGoursat[t/(t^3 - 1)^(1/3), t]` returns the obstruction $H_1 = z$, signalling non-elementarity (definitive in this case, since $F = t$ is polynomial).

The diagnostic test is purely algebraic and runs in time polynomial in the degrees of R and F (dominated by the cost of the rational-function arithmetic and root computation). Because no

symbolic-integration engine is invoked during the diagnostic phase, the “elementary” verdict is a constructive certificate: the rational integrand returned in the “**reductions**” (square-root case) or in the “**J*integrand**” fields (cube-root cases) makes the closed form computable by hand from elementary techniques (partial fractions, Euler substitution). Unlike the Risch–Trager algorithm, the package gives a direct geometric obstruction — the witnessing rational function on the cube-root cover — in the spirit of the constructive proofs of Theorems 3.3–4.3 and Corollary 4.9. For polynomial $F \in \mathbb{C}[t]$, the obstructed verdict is a complete certificate of non-elementarity (Theorem 4.3(iii) and Remark 4.4); for rational F with poles at non-branch points of the cube-root cover, the verdict identifies the obstructive eigencomponent but a complete decision requires the analysis of residues on Y_1 in the spirit of [16, 4]. By Corollary 4.9(iii), the same dichotomy on the full algebraic function field $\mathbb{C}(t, y)$, $y^3 = R(t)$, reduces to two cube-root sub-problems, each handled by the criteria above.

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